



# CoTune

Supervised by: Dr Mohamed Omar El-Shalakani

# Our Team



|                                          |         |
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# 01

## Introduction

Problem, Solution, Motivation

# 02

## Active Suspension Experiment

Overview, Mechanical Model,  
Dynamic Model, Sensors,  
State Space, Control,  
Simulink

# 03

## Inverted Pendulum Experiment

Overview, Mathematical  
Model, Parameter Estimation,  
Control, Simscape



# 01 ✨ Introduciton ✨

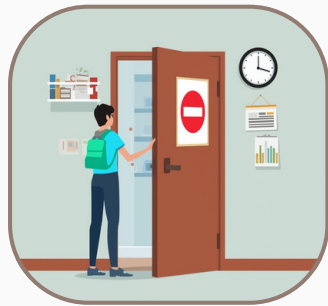


# The Problem with Traditional Labs



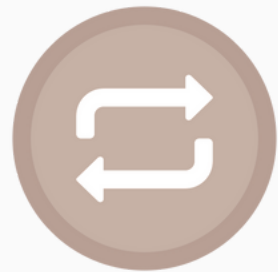
## Scheduling Constraints

- Limited experiment time



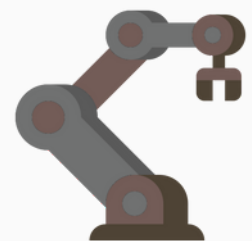
## Physical Presence

- Access is limited to on-site availability



## No continuous Experimentation

- Limited parameter tuning and iteration



## Restricted Hardware Availability

- Control theory stays mostly theoretical



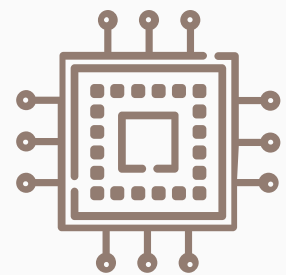


# Solution: CoTune Control Lab



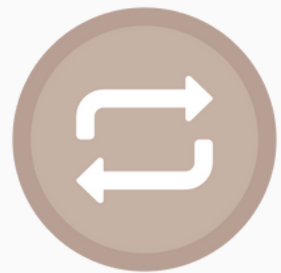
## Remote Access

- Access real lab from anywhere in real time.



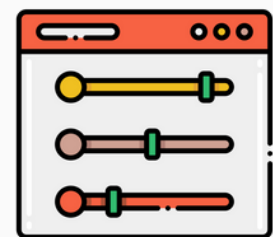
## Physical Hardware

- Experiment on actual physical laboratory equipment.



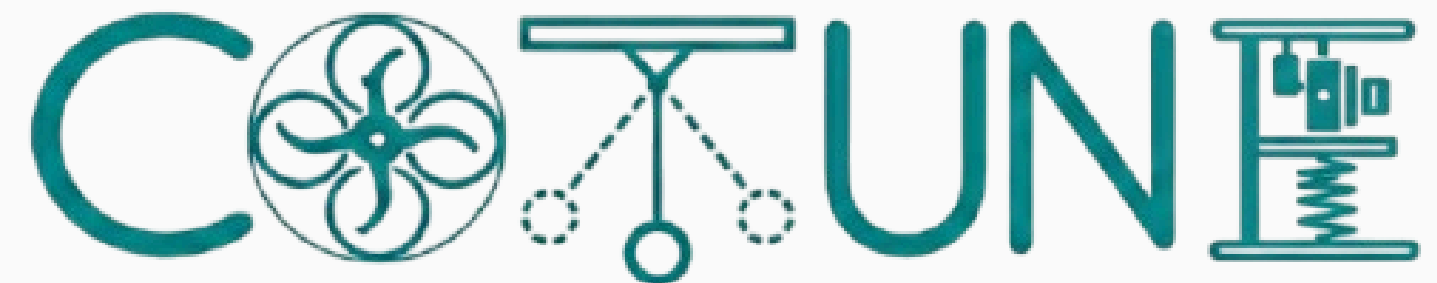
## Ongoing Tests

- Experiment endlessly without location or time limits.



## Parameter Tuning

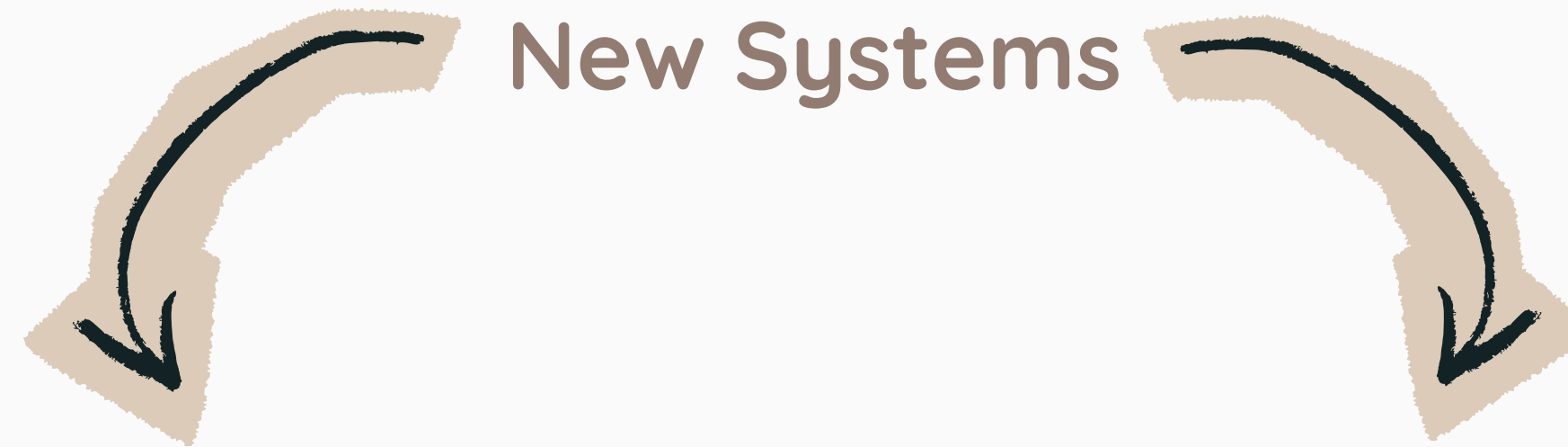
- Customize settings to optimize your experimental setup.





# Motivation for New Systems

CoTune Laboratory is expanding its capabilities by adding new physical systems to enhance educational experiences and practical insights.



## New Systems

### Active Suspension

Enables disturbance rejection.

### Rotary Inverted Pendulum

Demonstrates instability and real-time stabilization.



# 02 ★ Active Suspension ★



# Suspension System Overview

## Load Support

The suspension system supports vehicle weight while isolating the body from road disturbances for a smoother ride.

## Spring Mechanics

Springs absorb energy from road impacts, while dampers dissipate energy to control oscillations for vehicle stability.

## Tire Contact

Maintaining consistent tire-road contact is crucial for vehicle stability, enhancing traction and overall handling performance.

## Comfort vs. Handling

There is a fundamental trade-off between ride comfort and responsive handling, influencing suspension design and vehicle dynamics.

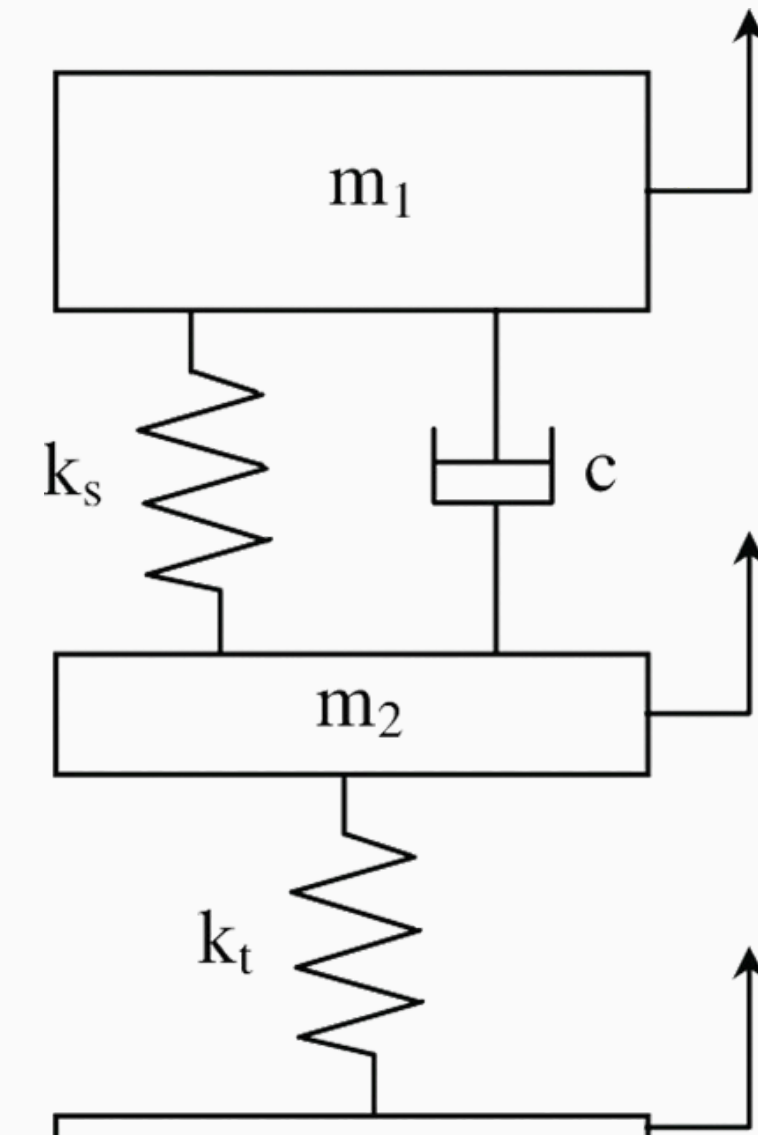




# Types of Suspension Systems

## PASSIVE SUSPENSIONS:

- Simple and cost-effective.
- They consist of a spring ( $k_s$ ) and a damper ( $c$ ) which are fixed parameters defined during the design phase.

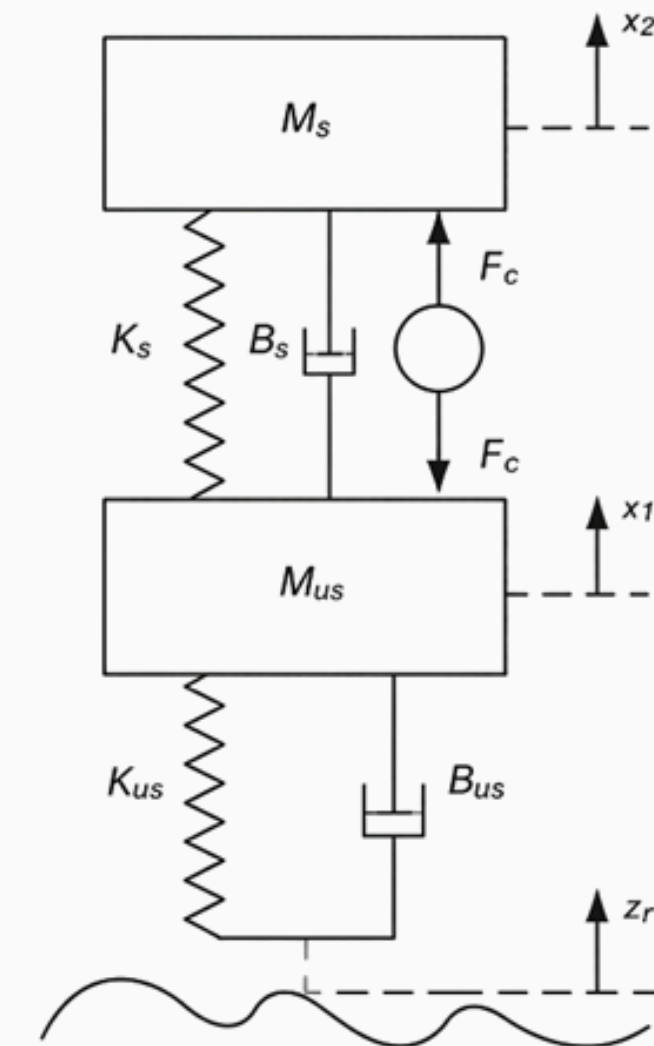




# Types of Suspension Systems

## ACTIVE SUSPENSIONS:

- Controlled actuation between vehicle body and wheel based on sensor feedback
- Real-time disturbance rejection instead of passive energy dissipation







# The Overall Mechanical Architecture



Inspired by Quanser Benchmark

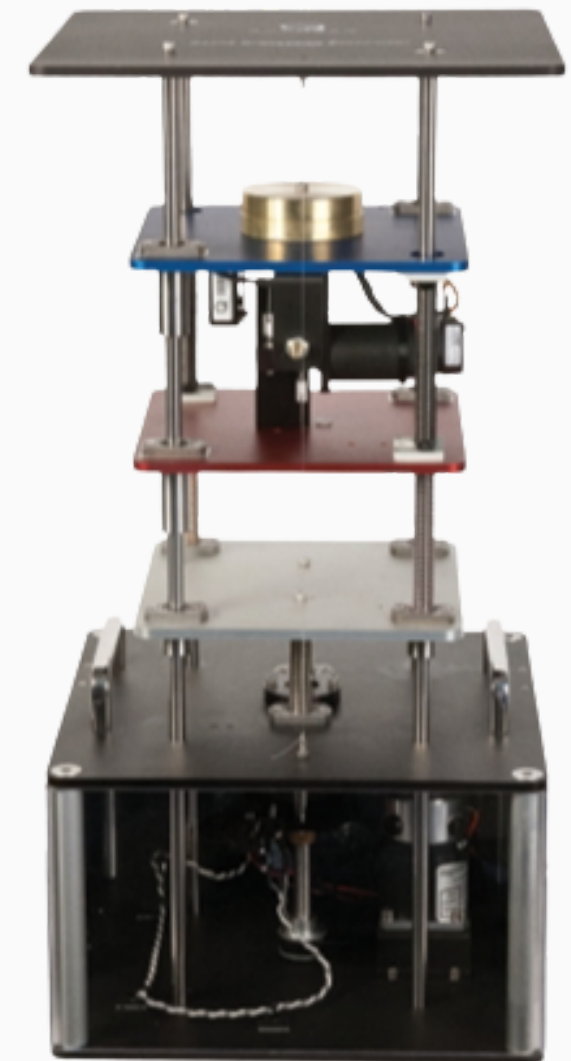
Top Plate — Sprung Mass (Vehicle Body)

Middle Plate — Unsprung Mass (Wheel Assembly)

Bottom Plate — Road Input (Base Excitation)

This arrangement allows the system to physically emulate the **classical quarter-car suspension model**, where:

- Relative displacement between the top and middle plates corresponds to suspension travel
- Relative displacement between the middle and bottom corresponds to tire deflection







# Assembly

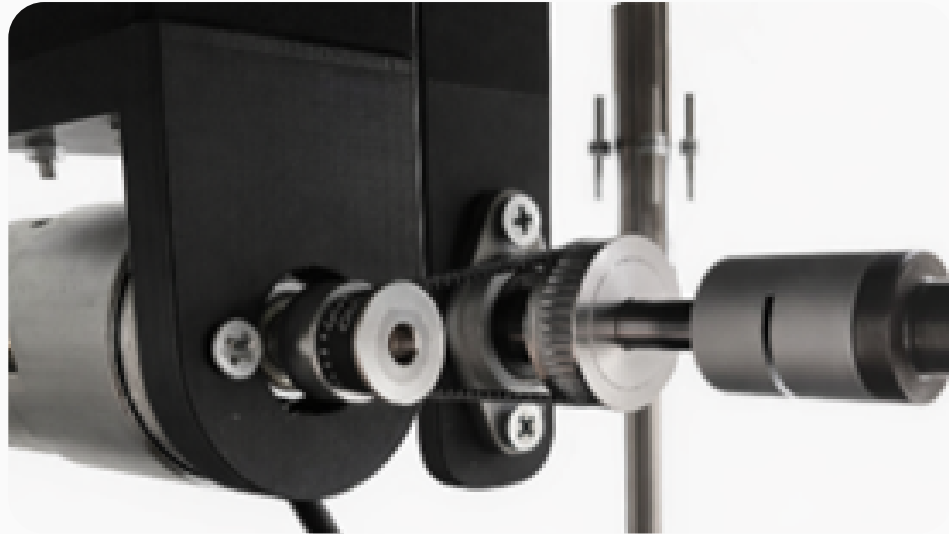


CAD



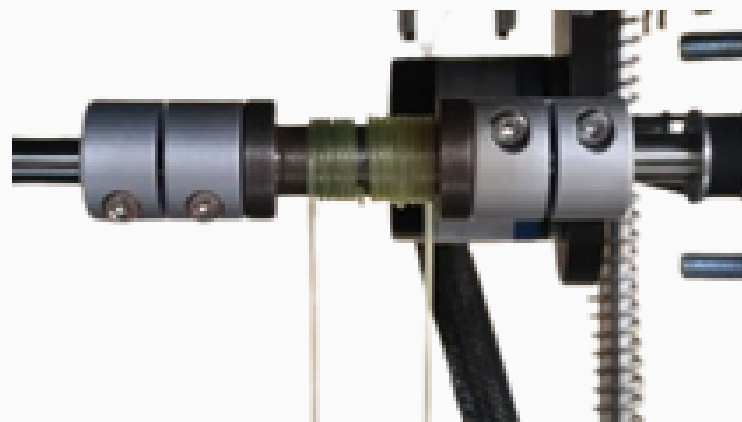
Real

# Capstan Mechanism



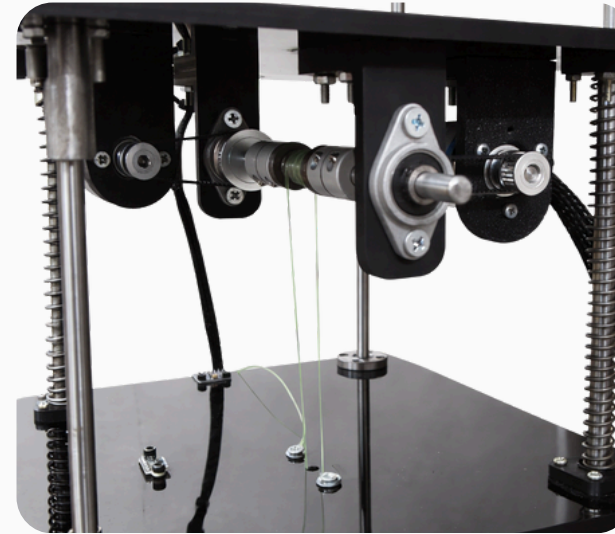
## 1 ROTARY MOTION

DC motor with a 2:1 reduction ratio drives capstan Pulley rotation.



## 2 CABLE-PULLEY TRANSMISSION

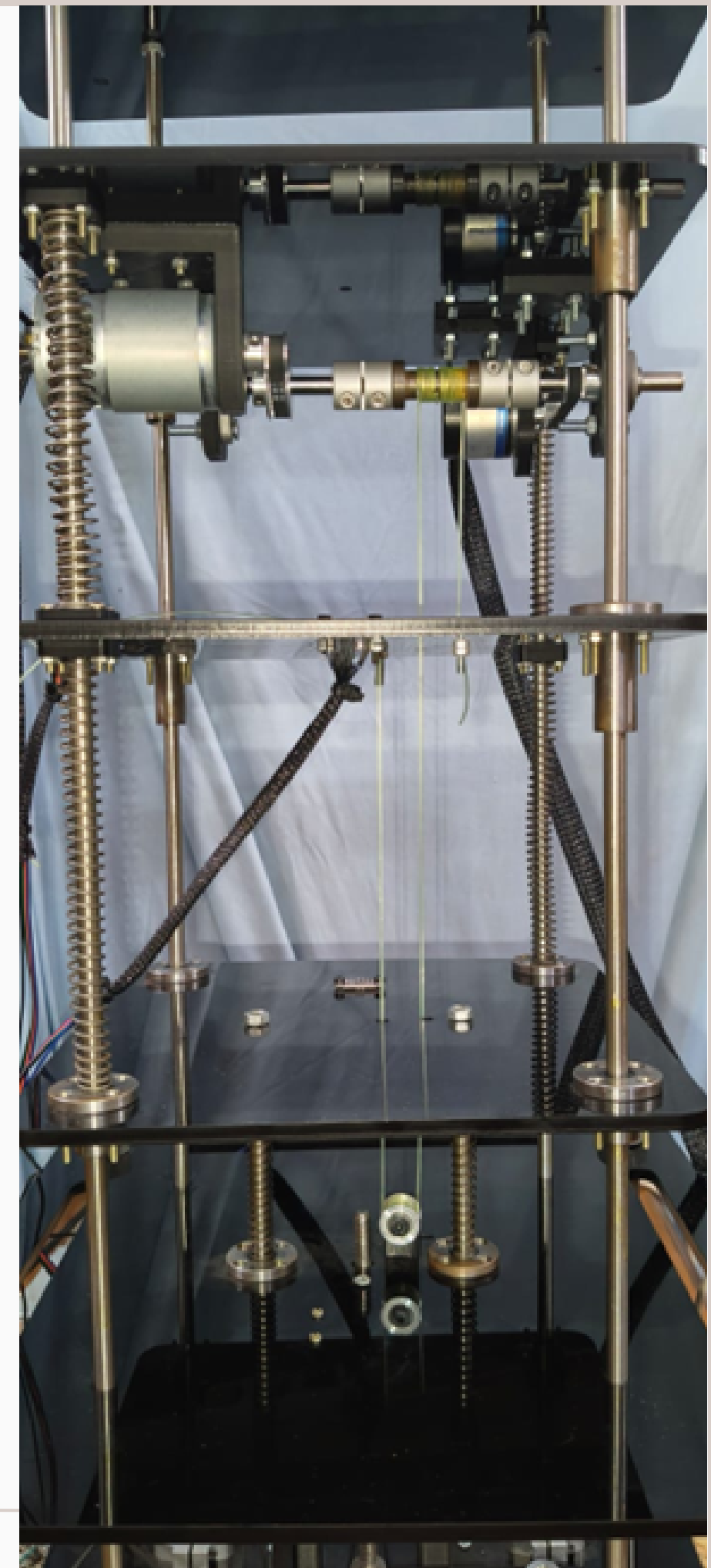
Cable wrapped around pulley converts rotation into motion with ( $r = 6\text{mm}$ )



## 3 Linear Force

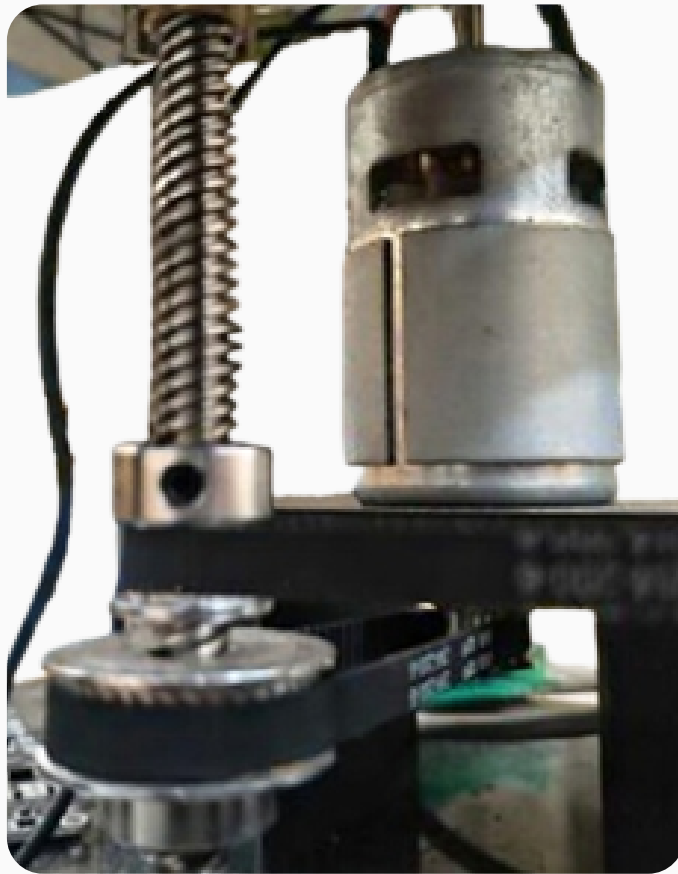
Cable tension generates upward / downward actuation.

## 4 SMOOTH LOW-BACKLASH ACTUATION



# Lead Screw Mechanism

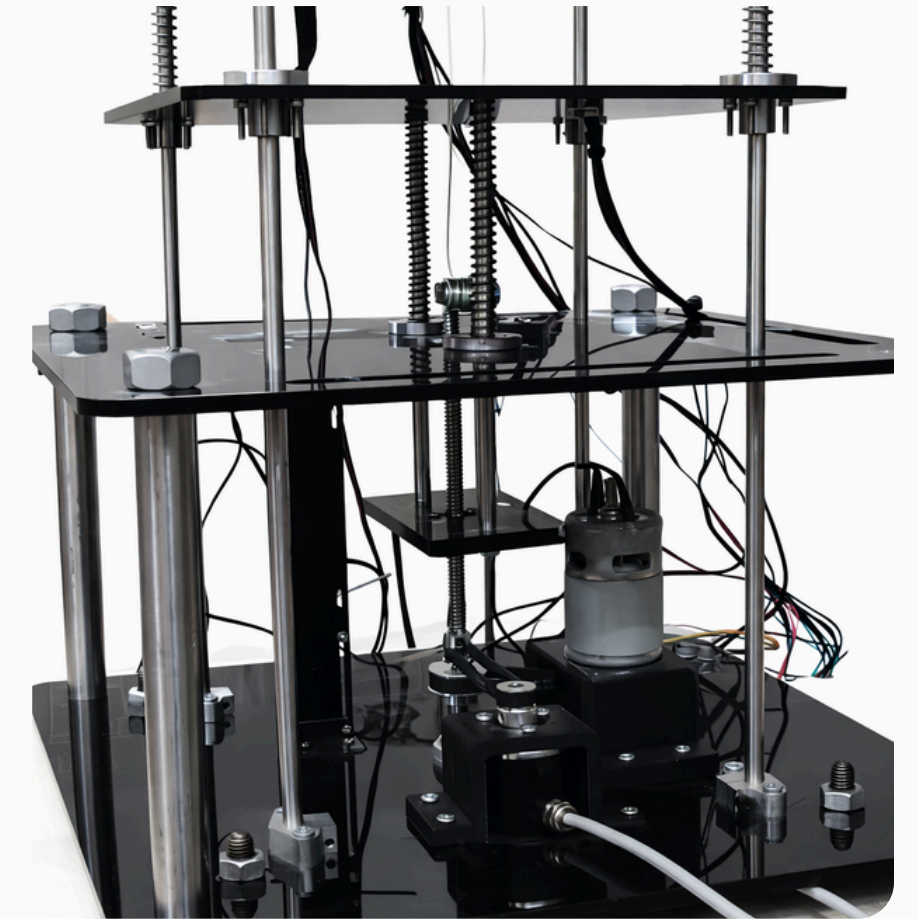
## For Road Excitation Actuator



- 1** **ROTARY MOTION**  
DC motor with a 2:1 Gear ratio  
drives Screw Rotation.



- 2** **LEAD SCREW TRANSMISSION**  
Nut converts rotation into linear motion  
(Pitch = 8 mm / rev)



- 3** **PRECISE VERTICAL DISPLACEMENT**  
Controlled base plate motion.

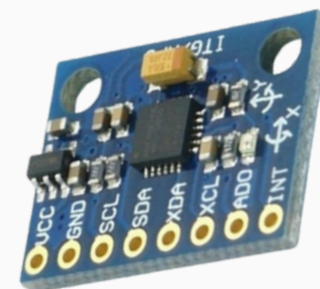
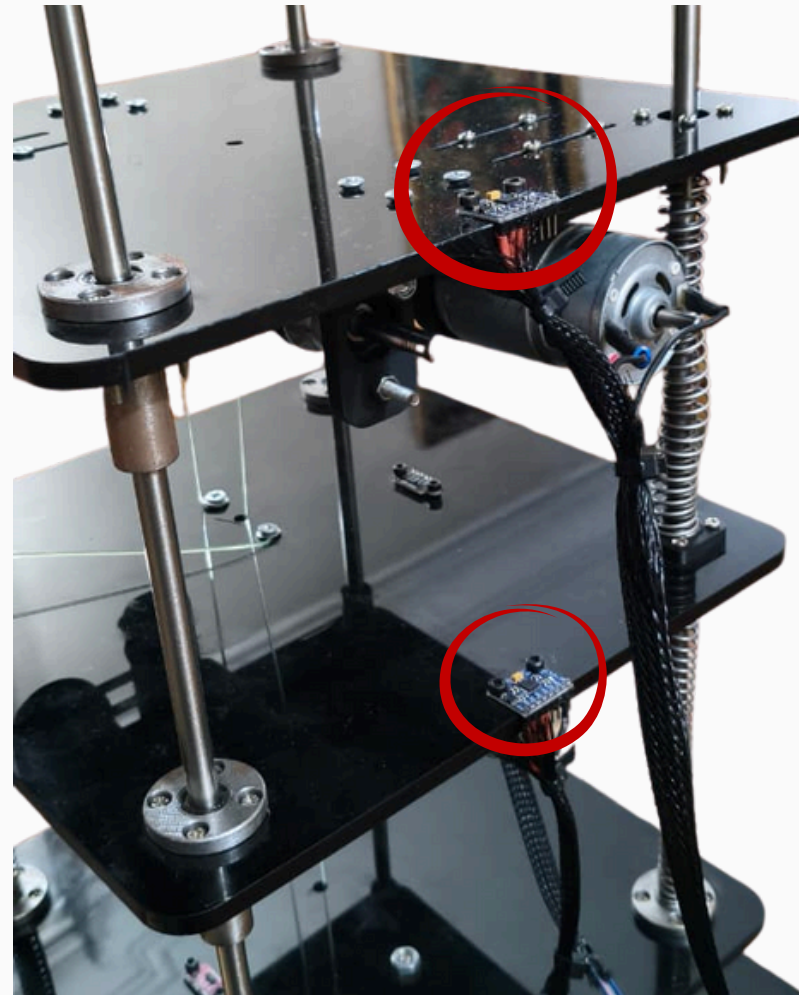


# Sensor Selection

## For Acceleration Measurement

Sensor 1 — Top Plate

Sensor 2 — Middle Plate



MPU6050

### Dynamic Response Measurement

Captures real-time vibration and ride comfort behavior.

### Compact Integrated IMU

3-axis accelerometer + gyroscope in a single module (I<sup>2</sup>C).

### Dual-Plate acceleration sensing

Vertical acceleration measured on sprung and unsprung masses.

### High-Resolution Configuration

Accelerometer range set to  $\pm 2$  g for small vibration sensing.

# Sensor Selection

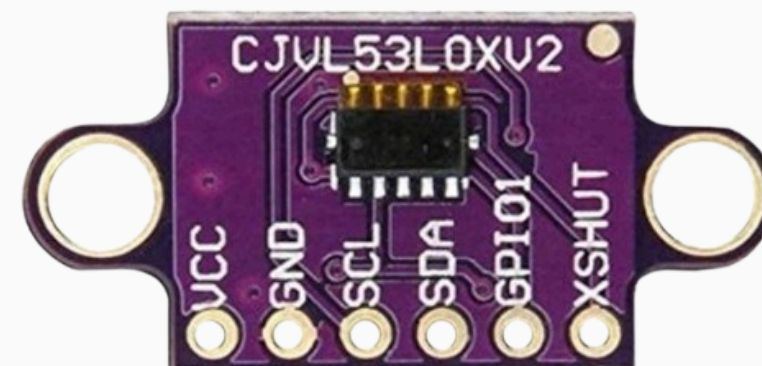
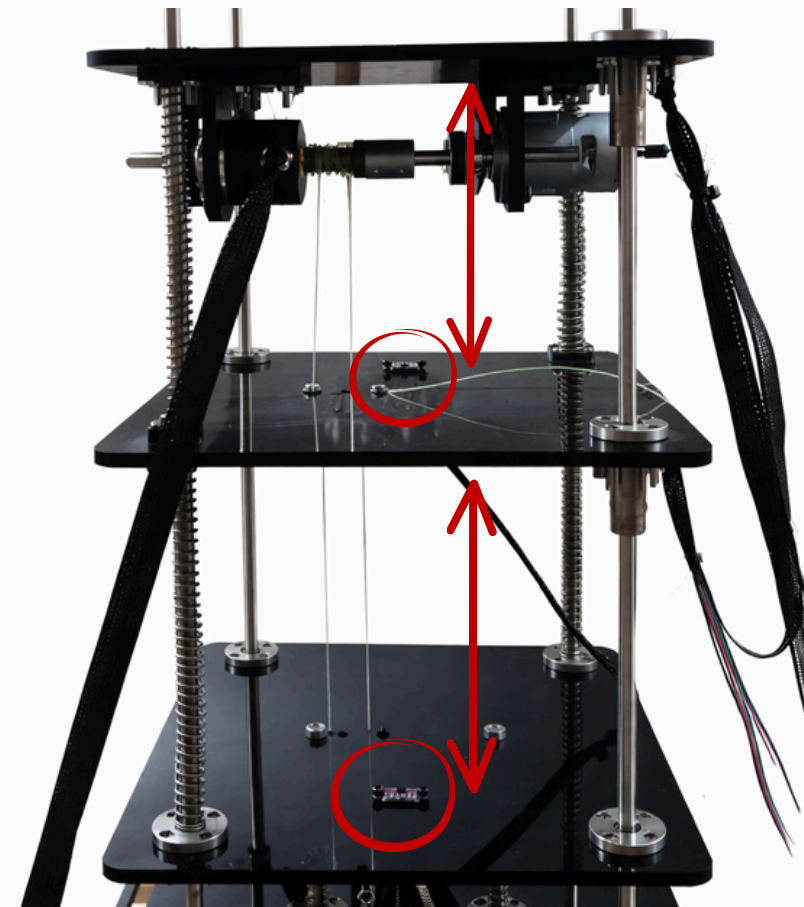
## For Displacement Measurement

Sensor 1 — Middle Plate

$$d1 = z_s - z_{us}$$

Sensor 2 — bottom Plate

$$d2 = z_{us} - z_r$$



VL53L0X

### Non-Contact Measurement

Two ToF sensors avoid friction and mass loading.

### Relative Displacement

Vertical relative displacement measured across suspension stages.

### Real-Time Measurement

Sufficient accuracy and sampling rate for closed-loop control.

### Easy Integration

Simple I2C interface for seamless system integration.



# Sensor Interfacing

## THE CHALLENGE: I2C ADDRESS CONFLICTS

- All sensors (ToF and MPU6050) communicate via a shared I2C bus at 3.3V.
- While MPU sensors have hardware pins to change addresses, the VL53L0X (ToF) sensors have a fixed default address.
- Using two identical ToF sensors simultaneously on one bus causes a collision, making standard initialization impossible.



# Sensor Interfacing

## THE SOLUTION: XSHUT SEQUENCING & S-FUNCTION

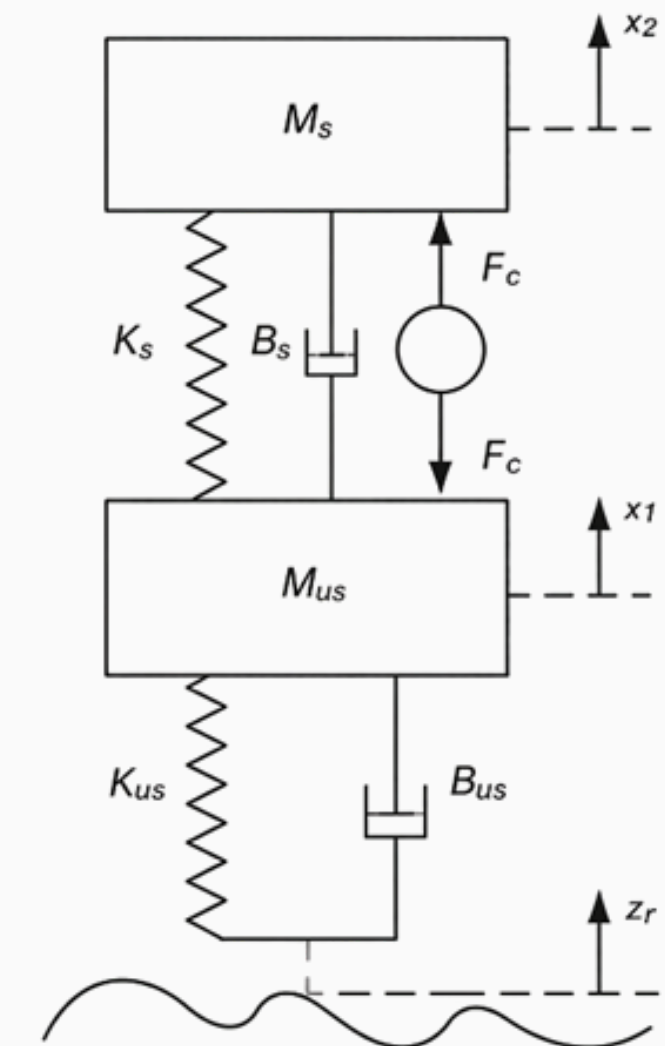
To resolve this, we implemented a sequential startup strategy using the XSHUT pin of each ToF sensor:

- Sequential Bring-up: The ESP32 triggers the XSHUT pins to enable each ToF sensor one at a time.
- Dynamic Reassignment: Each sensor is assigned a unique I2C address in software immediately upon "wake-up" to avoid bus collisions.
- Custom S-Function: A tailored C++ S-Function manages this timing-critical initialization, ensuring a deterministic startup before the control loop begins.

# Quarter-Car Suspension Model

## THE MODEL CONSISTS OF:

- Sprung mass:  $M_s$  [Kg]
- Unsprung mass:  $M_{us}$  [Kg]
- A suspension spring-damper pair ( $k_s, B_s$ ) connecting the sprung mass to the unsprung mass.
- A tire spring-damper pair ( $k_{us}, B_{us}$ ) connecting the unsprung mass to the road surface.
- The active suspension control command:  $F_c$  applied between the sprung and unsprung masses.
- A road displacement input  $z_r(t)$ .







# State Space Model

## Quarter-Car Dynamic Model



To optimize performance, we transform our 2-DOF differential equations into a state-space model:

$$\dot{x} = Ax + Bu$$

State Vector (x): Captures the four energy storage elements of the system.

- x1: Suspension Deflection ( $z_s - z_u$ )
- x2: Body Velocity ( $\dot{z}_s$ )
- x3: Tire Deflection ( $z_u - z_r$ )
- x4: Tire Velocity ( $\dot{z}_u$ )

Output Vector (y): Focused on measured suspension and tire deflection for real-time feedback.



# State Space Matrices

SYSTEM MATRIX (A)

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ -\frac{K_s}{M_s} & -\frac{B_s}{M_s} & 0 & \frac{B_s}{M_s} \\ 0 & 0 & 0 & 1 \\ \frac{K_s}{M_{us}} & \frac{B_s}{M_{us}} & -\frac{K_{us}}{M_{us}} & -\frac{(B_s+B_{us})}{M_{us}} \end{bmatrix}$$

INPUT MATRIX (B)

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & -1 \\ 0 & \frac{B_{us}}{M_{us}} \end{bmatrix}$$

OUTPUT MATRIX (C)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

FEEDTHROUGH MATRIX (D)

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



# Control Algorithm

## Quarter-Car Dynamic Model



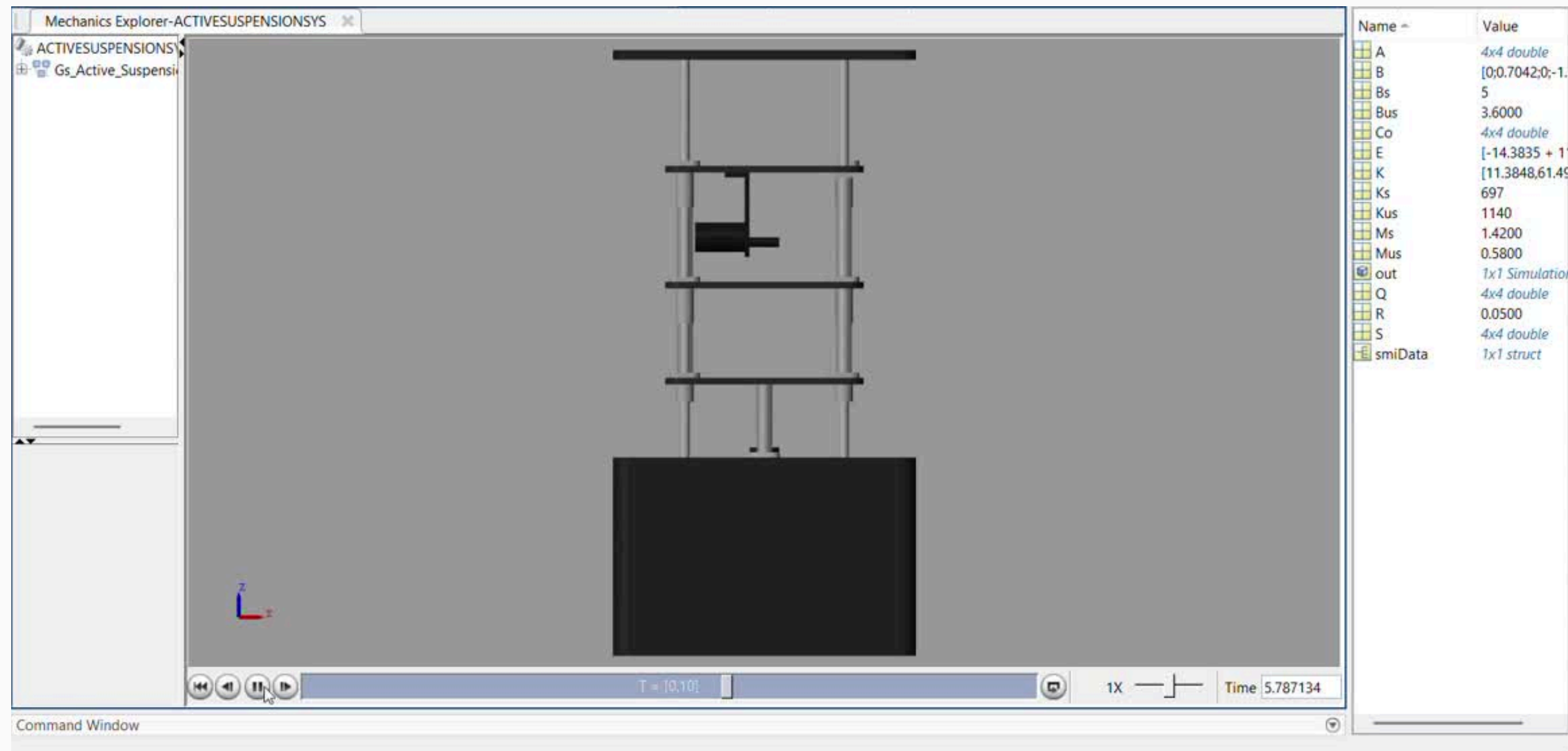
Rather than arbitrarily placing poles, which can lead to unrealistic actuator demands, we use LQR to minimize a quadratic cost function  $J$ :

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

- Physical Tuning ( $Q$  &  $R$ ): We assign weights to each state ( $Q$ ) and the control effort ( $R$ ). This allows us to balance "Passenger Comfort" against "Actuator Limits."



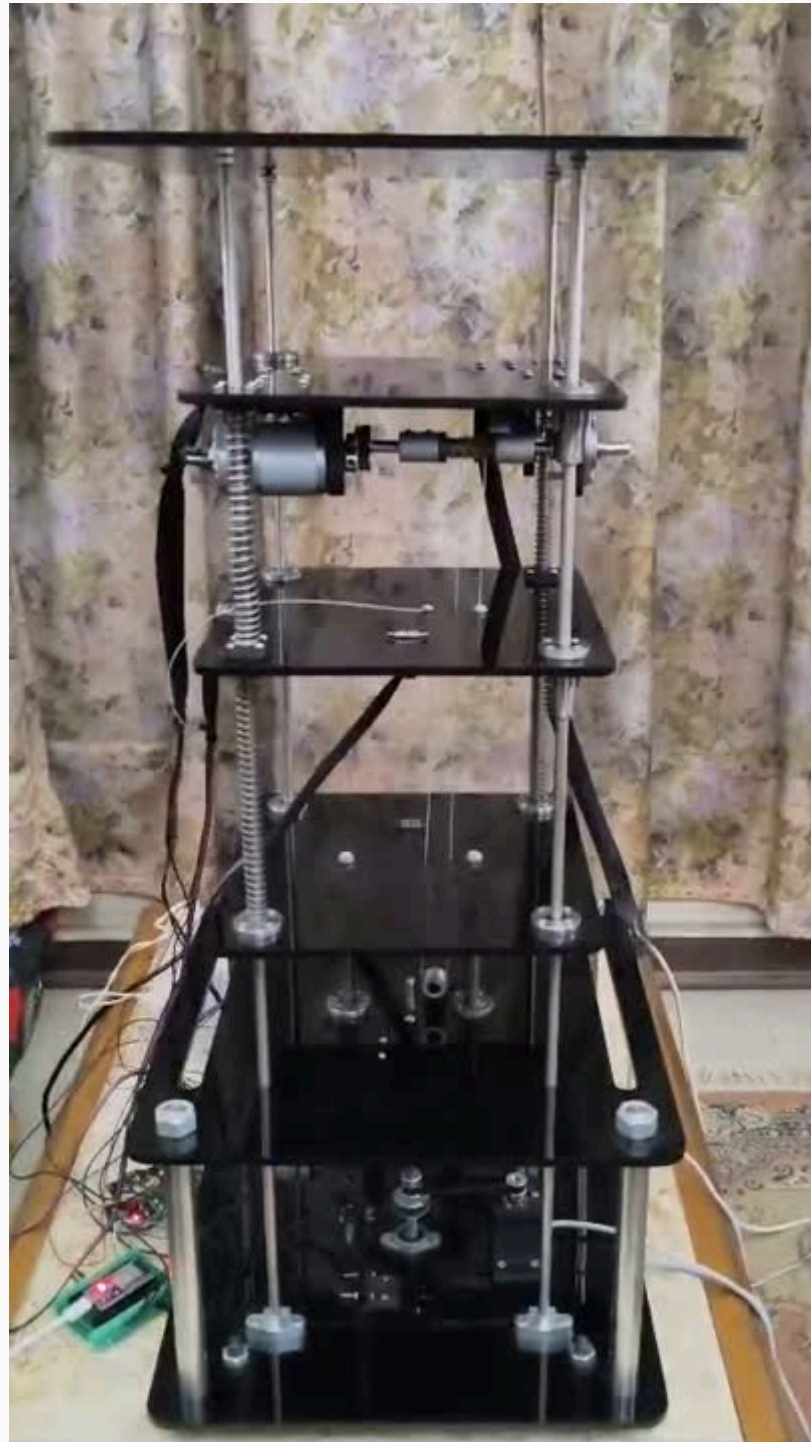
# Simscape Multibody Simulation



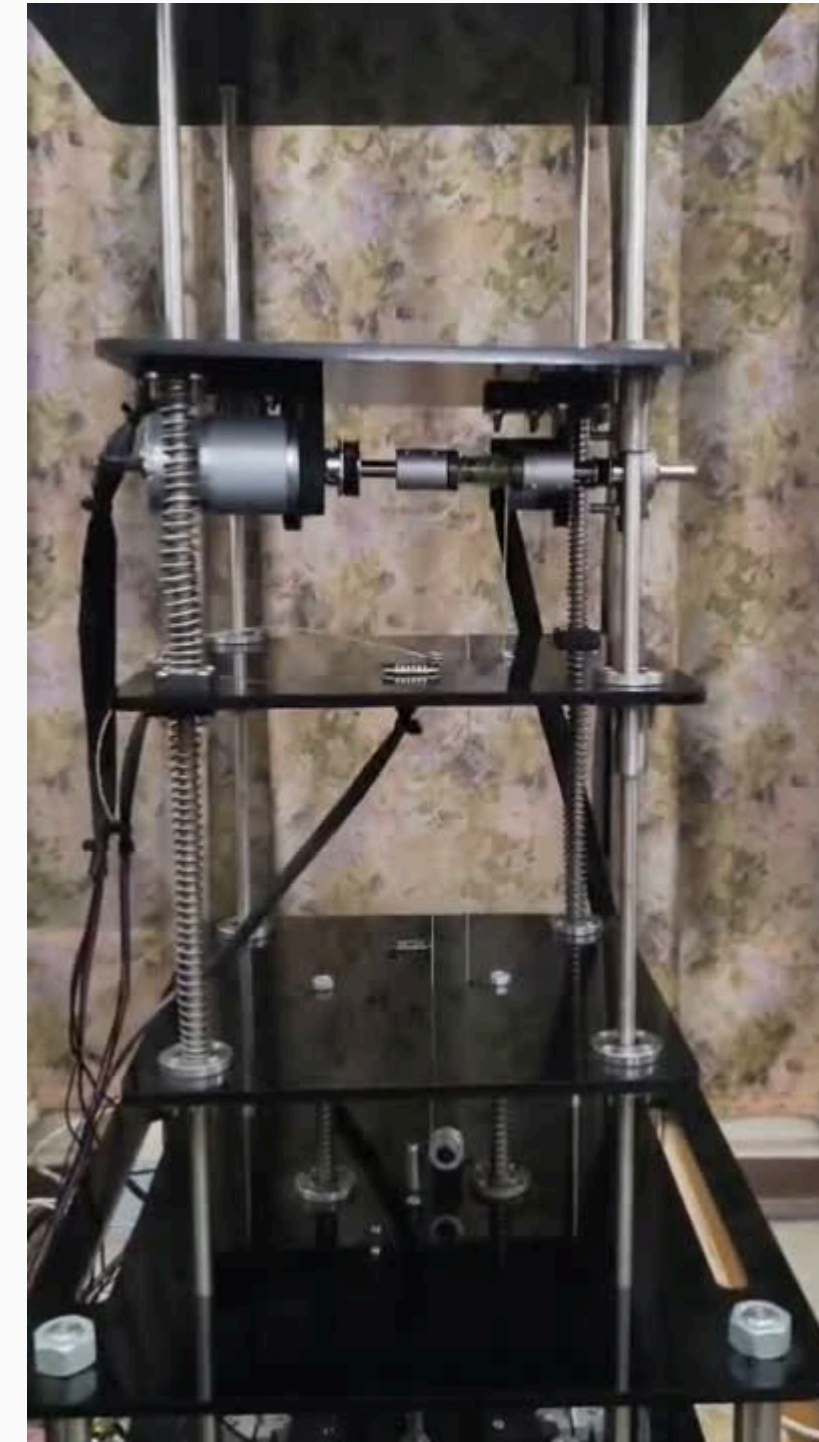


# Full System Demonstration

## Active Suspension



Road excitation sinusoidal input



Working capstan mechanism



# 03 ★ Rotary Inverted Pendulum ★





# what is an Inverted Pendulum

A PENDULUM THAT HAS ITS CENTER OF MASS ABOVE ITS PIVOT POINT.

- Inherently unstable.
- Must be actively balanced in order to remain upright.
- Must have a feedback system to keep it balanced.





# Types of Inverted Pendulum

IN GENERAL, TWO TYPES:

- Linear Inverted pendulum
- Rotary Inverted pendulum



Moving the pivot point horizontally



Applying a Torque at the pivot point



# The Overall Mechanical Architecture



Inspired by Quanser Benchmark

## ROTARY ARM—ACTUATED AXIS

- Horizontally rotating arm driven about a fixed vertical axis.

## PENDULUM LINK— FREE ROTATION

- Pendulum rotates freely in the vertical plane.

## UNDERACTUATED SYSTEM

- Two coupled degrees of freedom with single actuation.

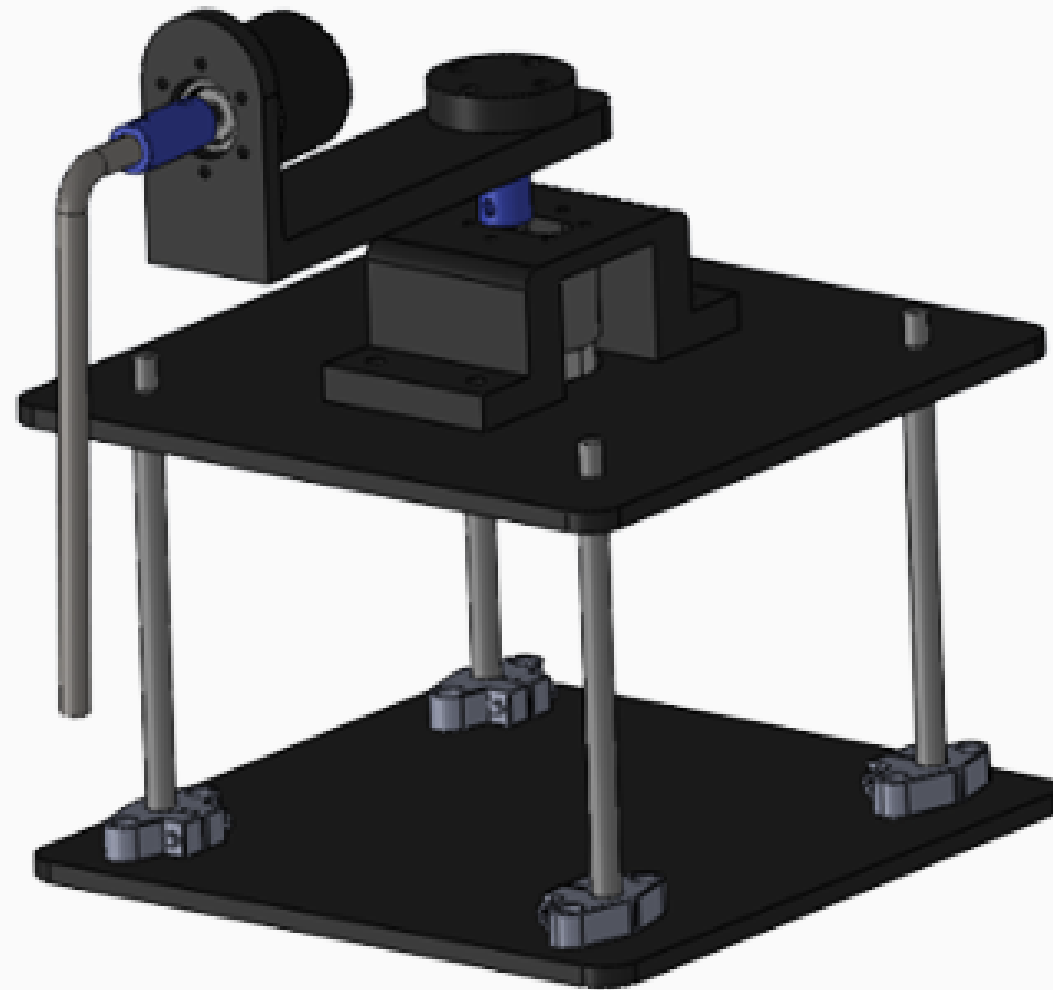
## UNSTABLE UPRIGHT EQUILIBRIUM

- Upright position represents an unstable operating point.

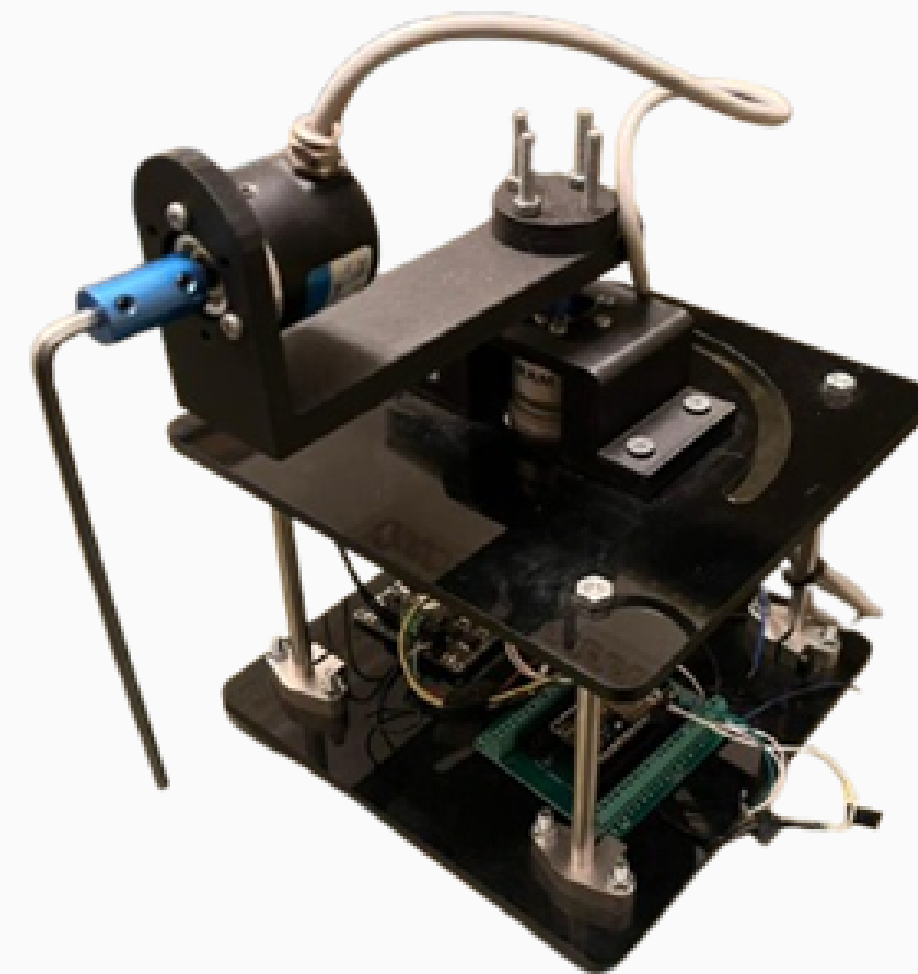




# Assembly



CAD



Real



# Mathematical Model

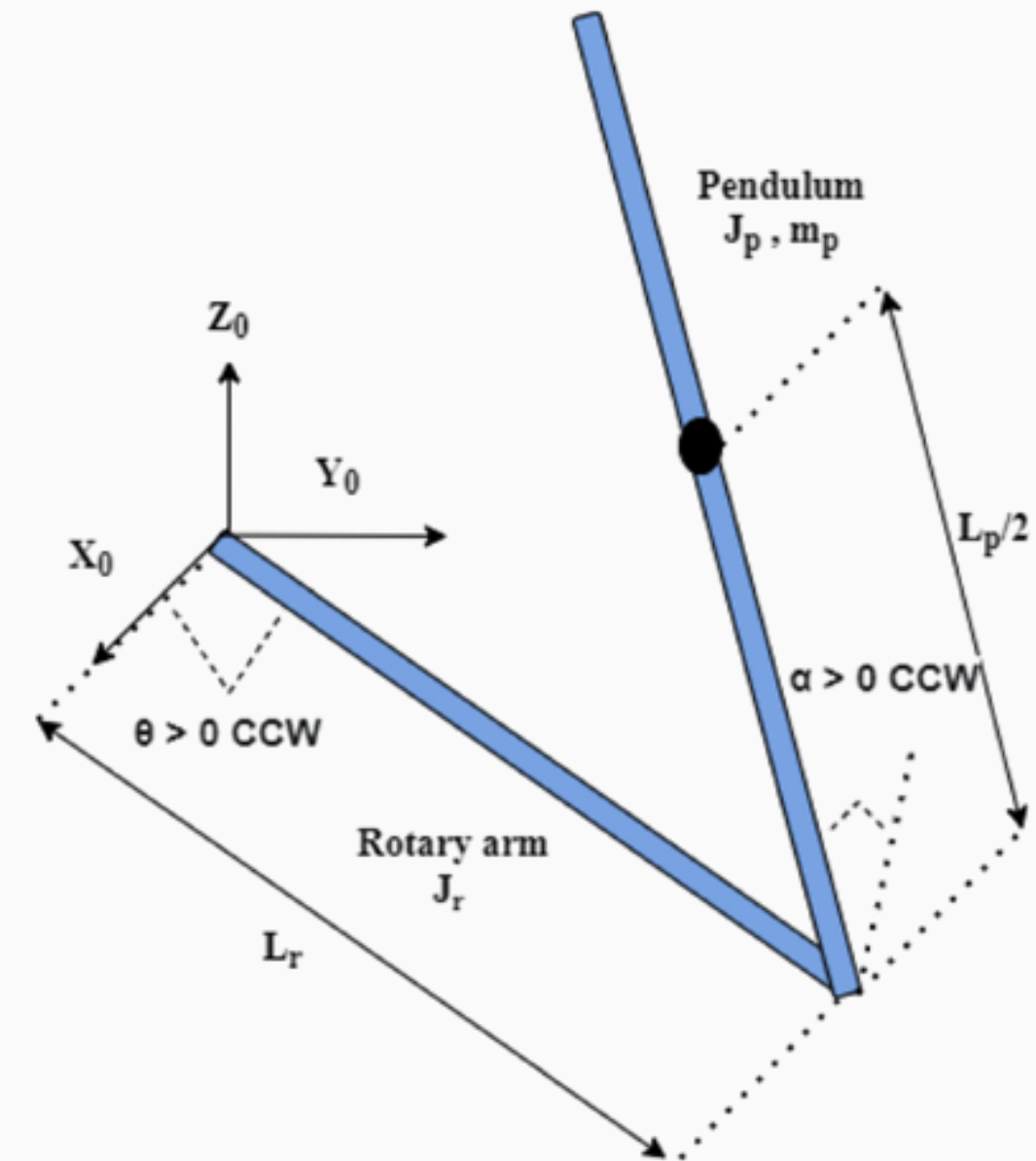
Equation of Motion



Linearization



State Space Model





# Motor Parameter Estimation

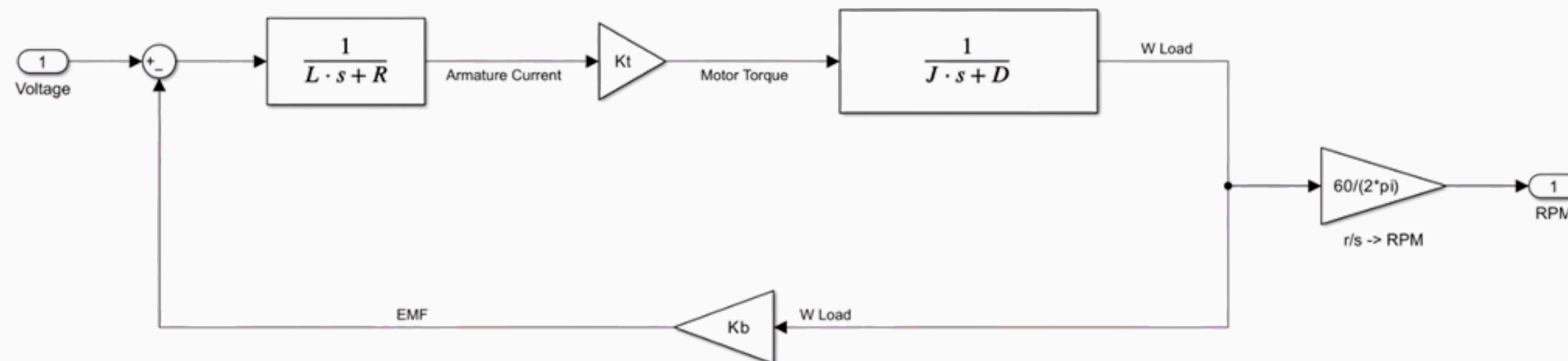


## WHY ESTIMATION

- Accurate actuator dynamics for LQR design

## ESTIMATED PARAMETERS

- Armature resistance  $R$
- Armature inductance  $L$
- Torque constant  $K_t$
- Rotor inertia  $J$
- Viscous damping  $D$



Motor Model



# System Parameter Estimation

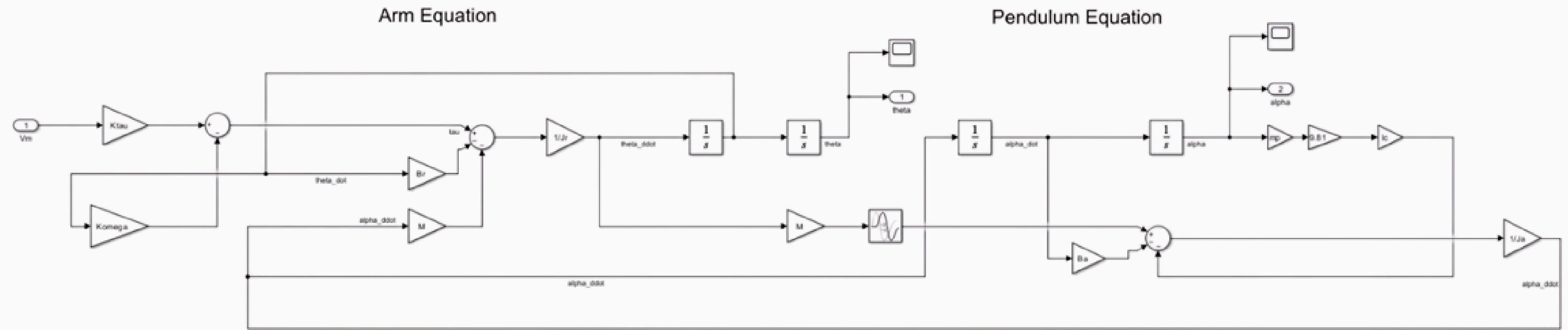


## ESTIMATION STRATEGY

- Motor parameters fixed from Stage 1
- Full arm-pendulum dynamic model used

## ESTIMATED PARAMETERS

- Rotary arm inertia  $J_r$
- Pendulum inertia  $J_p$
- Rotary arm viscous damping  $B_r$
- Pendulum viscous damping  $B_p$



System Linear Model



# Control Strategy

Essential strategies for rotary systems

## 1 SWING-UP CONTROL

A nonlinear control law is used to transfer the pendulum from its stable downward equilibrium to a region close to the upright configuration.

## 2 STABILIZATION CONTROL

Once the pendulum reaches the vicinity of the upright position, a linear controller is activated to stabilize and maintain the pendulum around this unstable equilibrium.



# Swing-Up Control

## Rotary Inverted Pendulum

### 1 METHOD

Energy-based swing-up control to elevate the pendulum.

### 2 ACTUATION

Motor torque applied through rotary arm for movement.

### 3 MODE SWITCHING

Automatic transition to LQR near upright position.

A switching condition was applied:  $|\alpha| < 15^\circ \rightarrow$  Switch to stabilization controller





# Stabilization Using LQR

## Upright Stabilization Using Optimal State Feedback



The resulting optimal state-feedback control law is given by:

$$u = -Kx$$

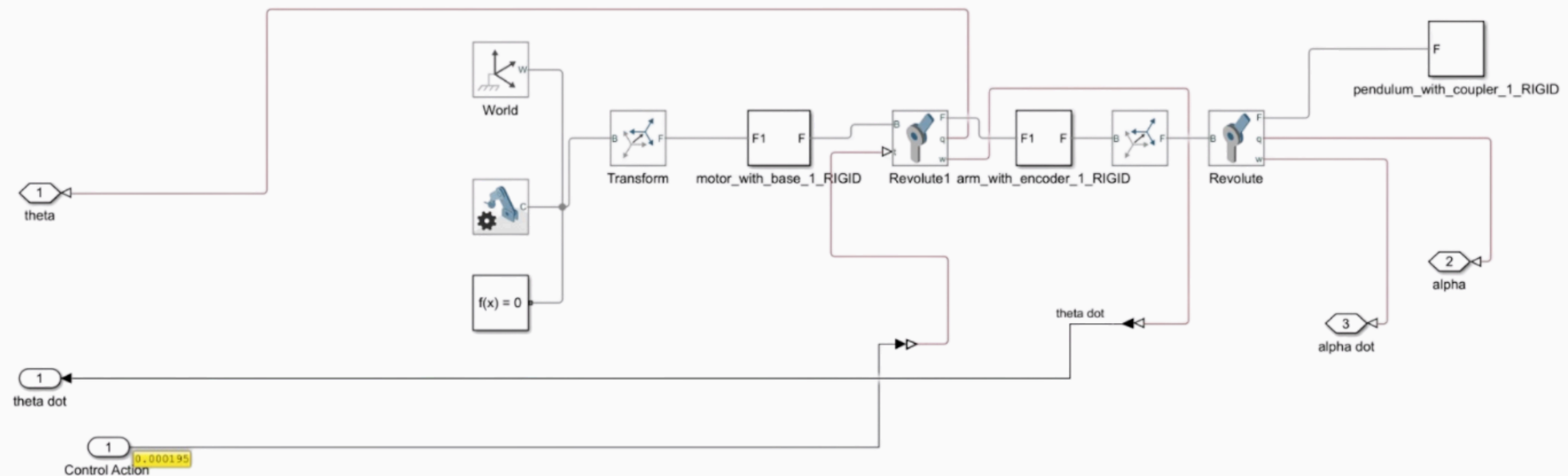
The system state vector is defined as:

$$x = [\theta \quad \alpha \quad \dot{\theta} \quad \dot{\alpha}]^T$$

- $\theta$  is the rotary arm angle,
- $\alpha$  is the pendulum angle measured from the upright position
- $\dot{\theta}$  is the angular velocity of the rotary arm,
- $\dot{\alpha}$  is the angular velocity of the pendulum.

# Simscape Multibody Physical Model

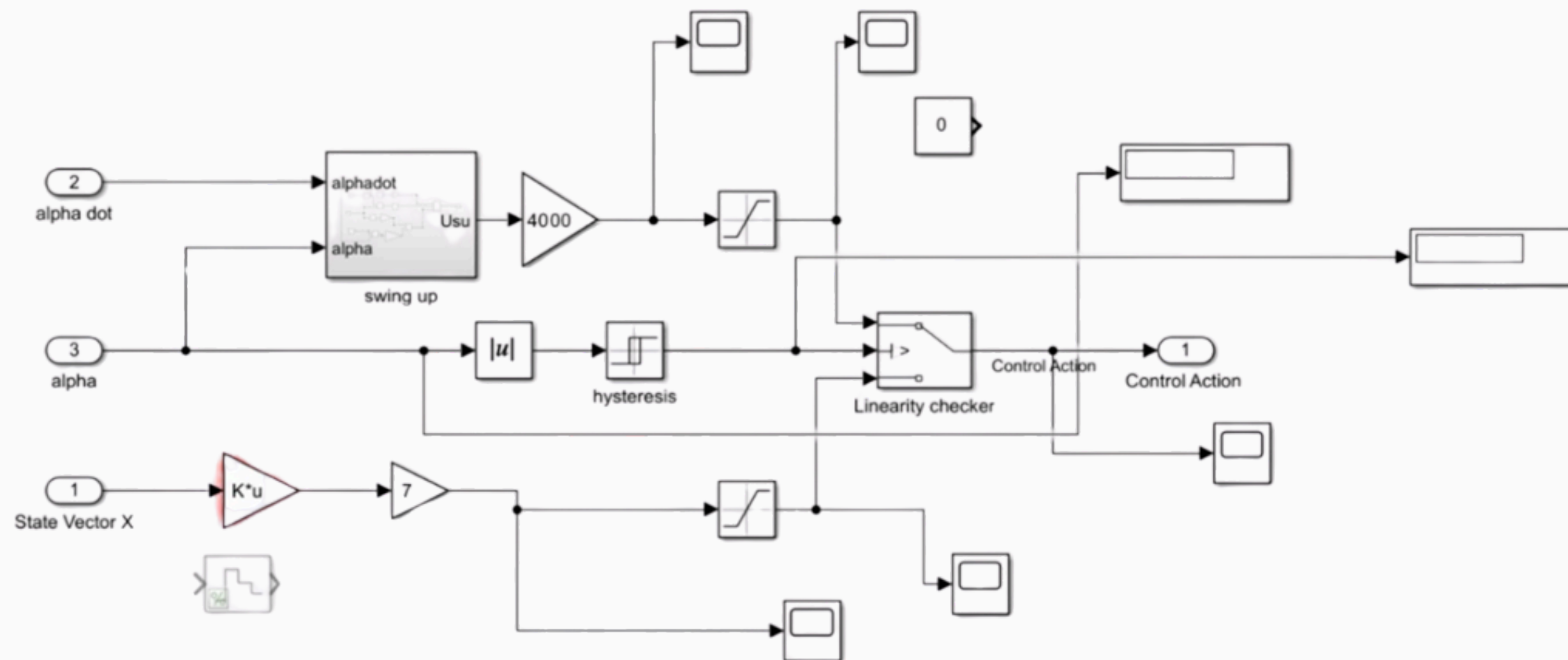
Imported Mechanical CAD into Simscape Multibody





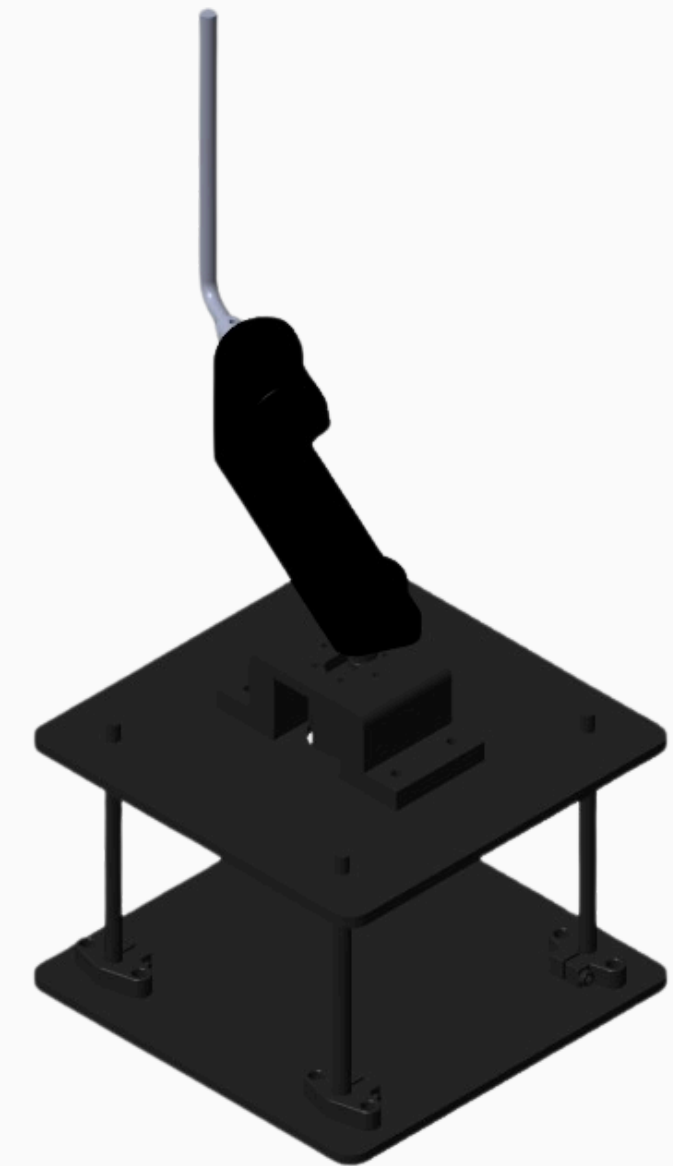
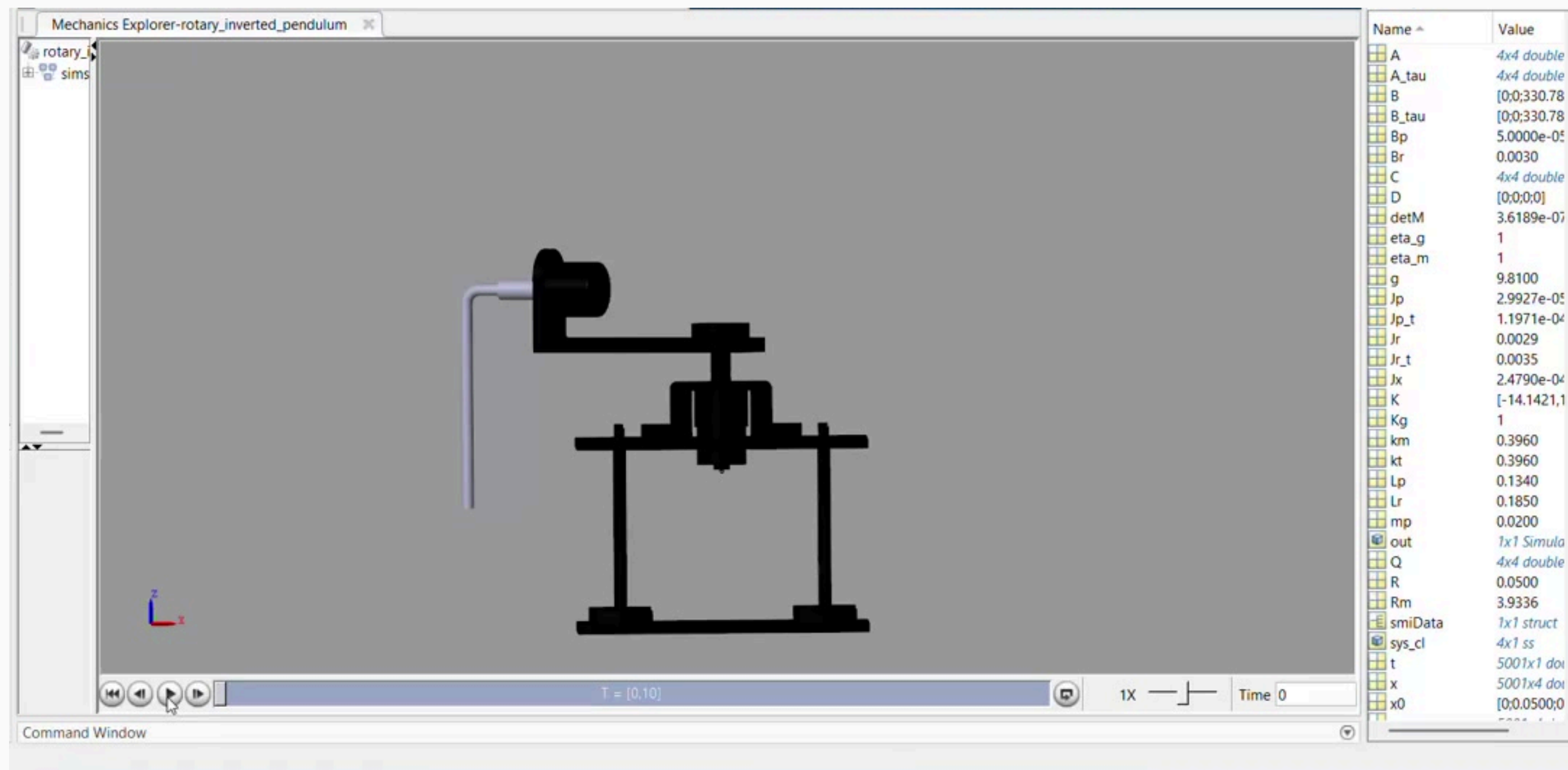
# Simscape Multibody Physical Model

Then add the LQR and swing up controller



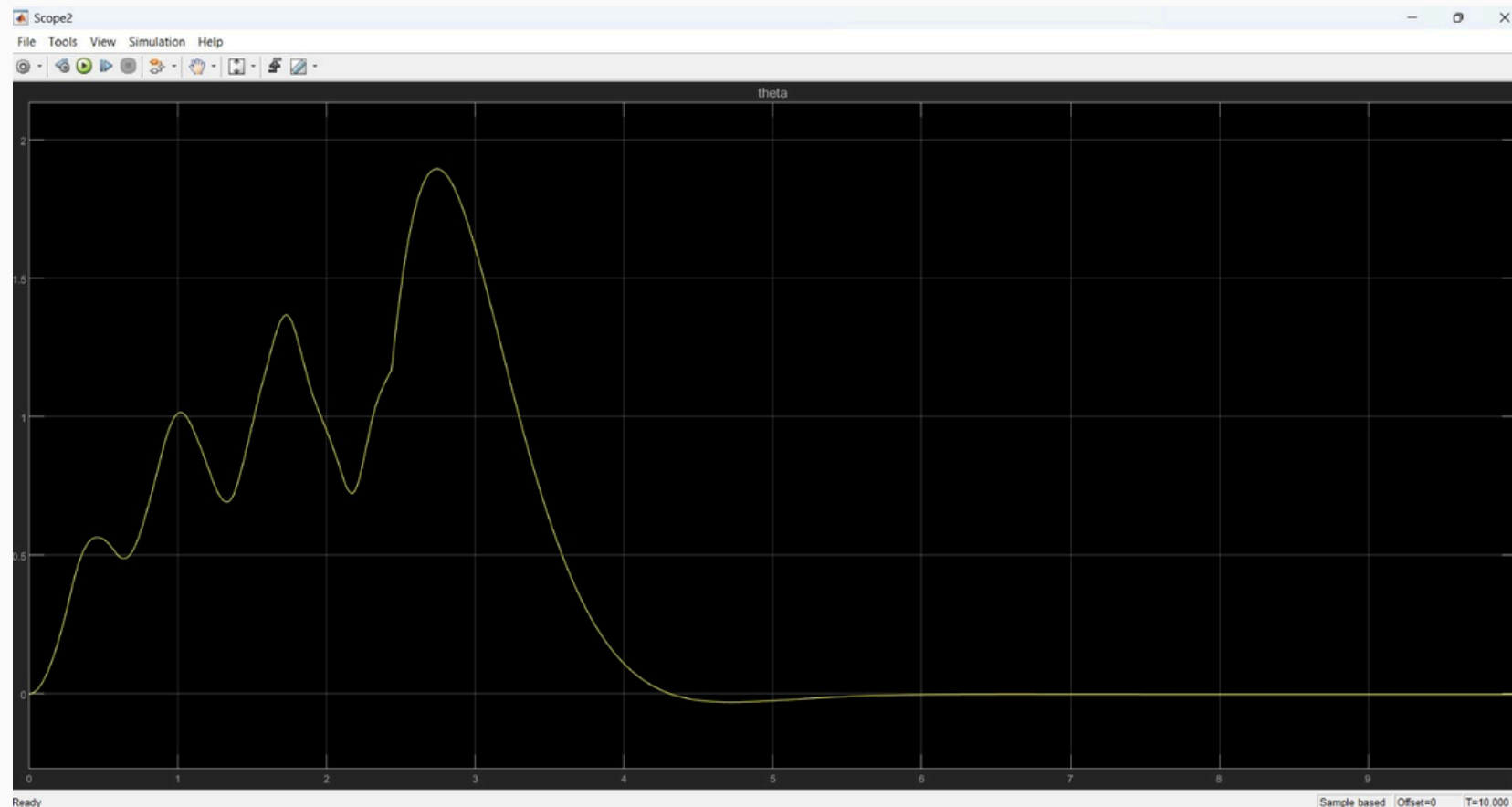


# Simscape Multibody Physical Model Simulation

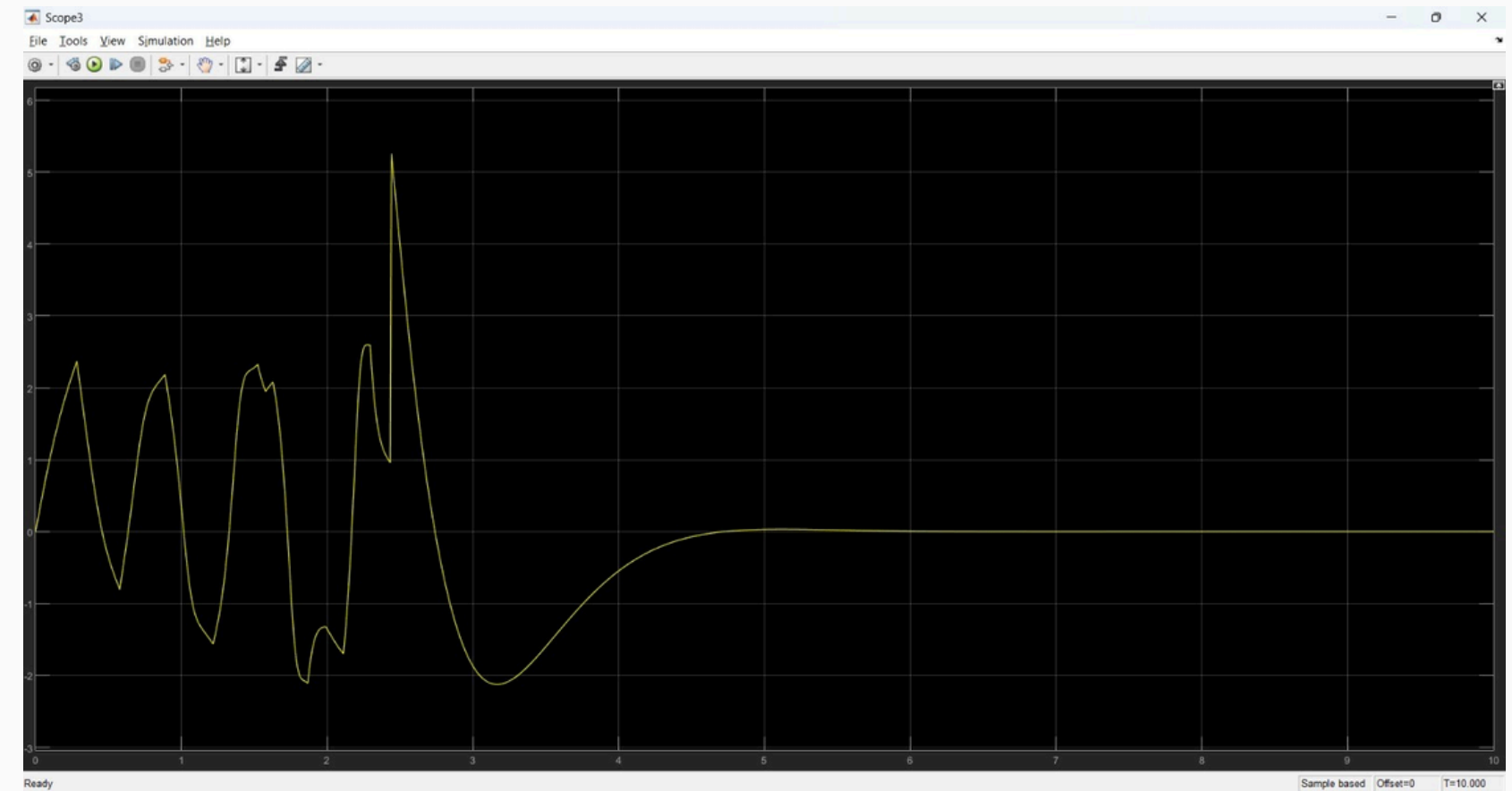


# Simscape Multibody Physical Model

## Simulation Results



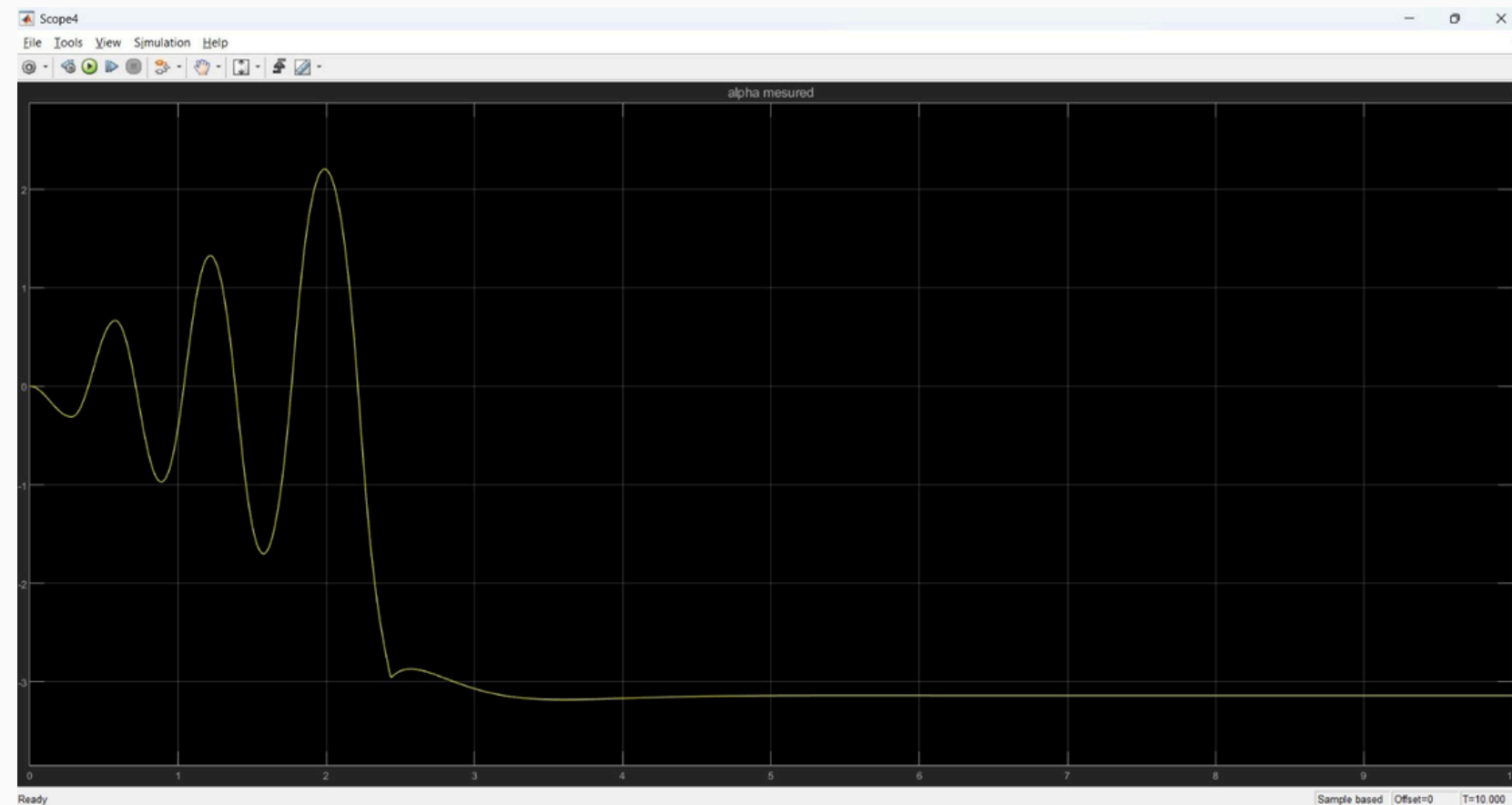
Theta



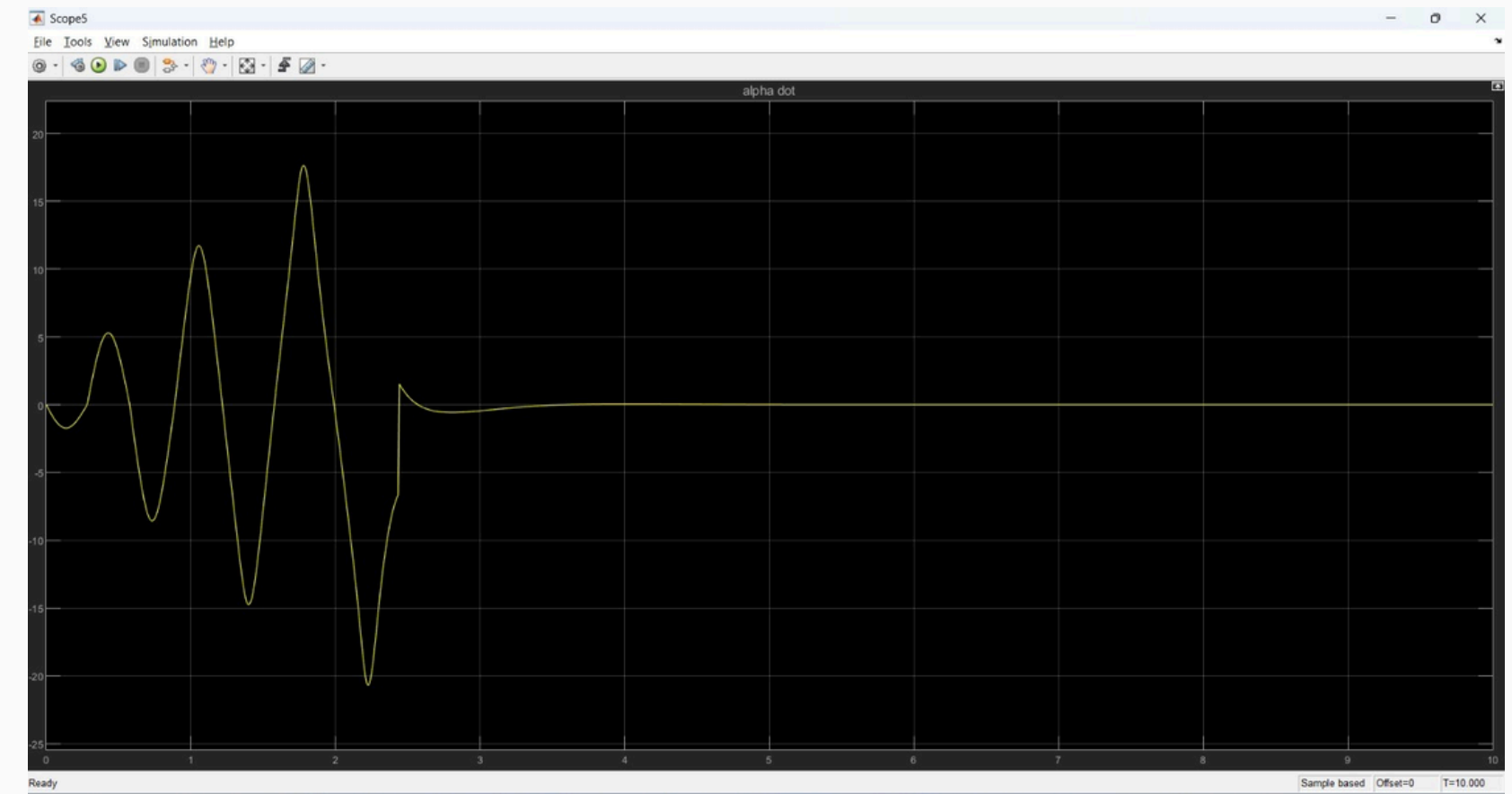
Theta dot

# Simscape Multibody Physical Model

## Simulation Results



Alpha

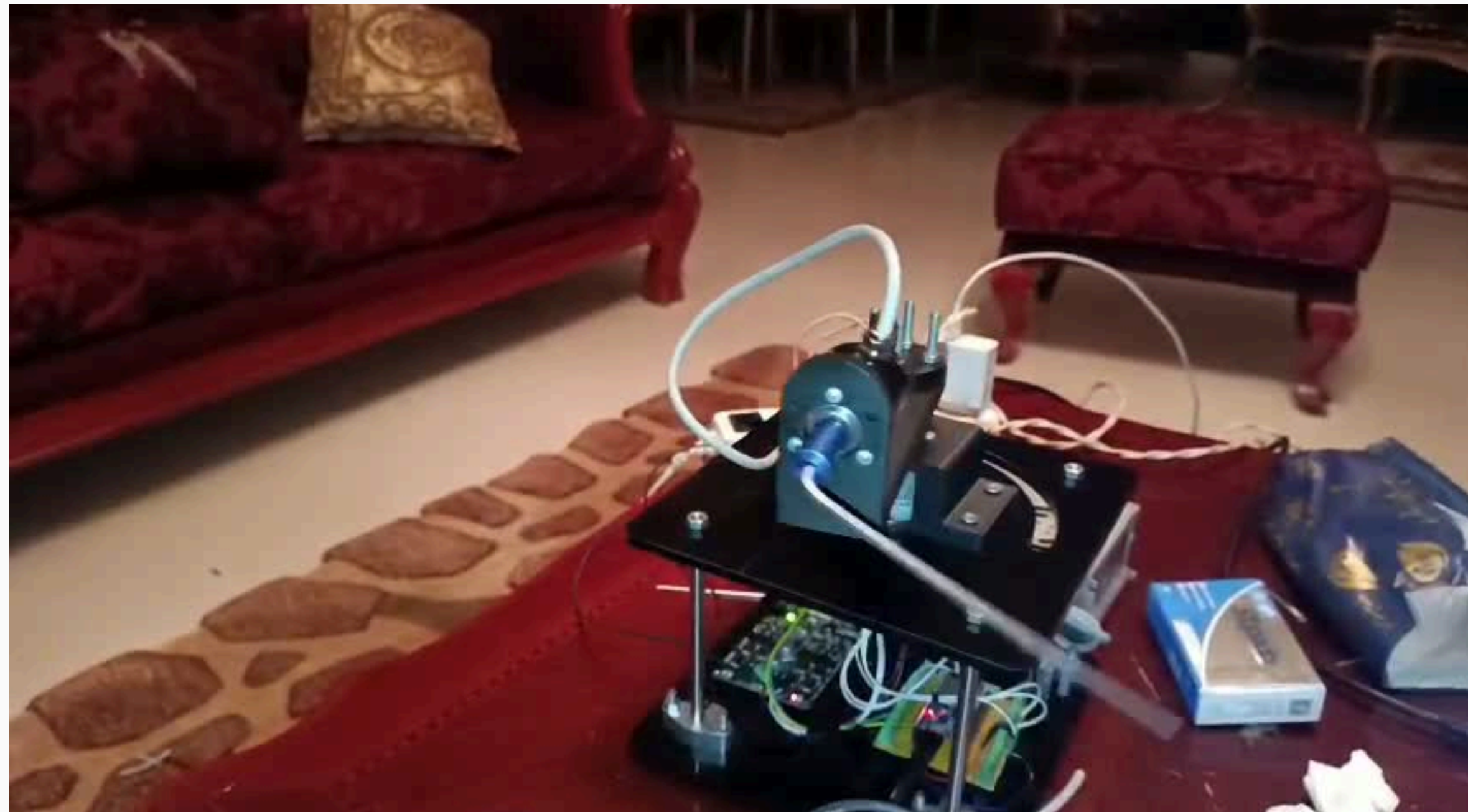


Alpha dot



# Full System Demonstration

## Rotary Inverted Pendulum





# Conclusion

This project successfully developed and validated two modular, high-fidelity experimental platforms designed to bridge the gap between theoretical control and real-world hardware.

## Active Suspension System

- Framework: Developed a complete mechanical and electrical architecture using a unified SolidWorks–Simscape workflow.
- Performance: Simulation of the LQR controller demonstrated significant improvements in ride comfort, road handling, and suspension travel over passive systems.





# Conclusion

## Rotary Inverted Pendulum

- Achieved successful real-time stabilization on the physical plant, confirming high consistency between the dynamic model and experimental behavior.
- Robust hands-on platform for studying unstable systems and real-time control strategies.

## Project Impact

- Integrated mechanical design, electrical interfacing, and advanced control within a single educational ecosystem.
- Both platforms are fully prepared for integration into the CoTune remote-access laboratory, supporting flexible and continuous engineering education.



# Practical Challenges

## 2. Capstan Mechanism Tension Loss

- Issue: String slippage and loss of tension during rapid downward movements.
- Root Cause: Insufficient mechanical engagement between the cable & the motor pulley.
- Solution: Redesigned the pulley with a internal guiding hole. Routing the string through the pulley core ensured positive engagement and eliminated slippage.





# Practical Challenges

## 3. Quantization in Rotary Pendulum

- Issue: Extremely low encoder resolution (7 PPR) on the rotary arm motor.
- Impact: Coarse angular steps ( $\sim 3^\circ$ ) made it impossible to detect the fine deviations required for upright stabilization.
- Planned Mitigation: Replacing the existing motor assembly with a high-resolution optical encoder to provide the precision needed for LQR balancing.

# Future Road map

## SENSOR FUSION

Kalman Filter Integration to Fuse ToF and IMU data for accurate state estimation. (week 1-2)

## CLOUD MPC (CNMPC)

Remote Optimization Control  
Implement MPC via CoTune cloud interface. (week 6)

## RL TUTOR

Reinforcement Learning Module  
Compare RL performance against LQR and MPC. (week 12)

01

02

03

04

05

## HARDWARE DEPLOYMENT

Real-Time Control Implementation  
Deploy controllers on physical suspension hardware. (week 2)

## DIGITAL TWIN

Live Simulation Synchronization  
Real-time data linked with simulation models. (week 10)



✧ Thank You! ✧

